Week 7 cargo cradle build card

Team ____ Builders A ____ B ____ C ____

Job: Carry one lightweight supply cube quickly without dropping it. Keep the shared robot assembled. Build a removable cradle that fits the mentor's mounting template.

Materials: Your team pack of spare beams/pins or pre-cut card, two rail strips, two brace strips, tape, one cube, and a footprint/mounting proxy. The robot stays with the other team while you build.

Build and try

1. Put the cube on your proxy. Choose a low mounting position within the marked area. Leave the wheels, cables and buttons clear.

2. Make a flat platform from card or spare parts. It must fit inside the approved mounting outline. Do not remove any part of the robot's chassis.

3. Add two low side rails. With card, fold a strip along its length and tape one side to the platform. Keep a little space around the cube so you can load it easily.

4. Add an end stop and brace a sagging section. Use the available strips or fold the platform edge. Avoid a large, heavy tower.

5. On the proxy, gently tilt forward, backward and sideways. Watch for sliding cargo, loose joints or a sagging rail. Repeat after changing one feature.

6. Show the mentor your change and fit the cradle only after the robot stops and the mentor checks it. At the robot, compare it with the mentor tray at the **same speed**.

Three jobs

Builder connects pieces. Designer chooses parts and checks fit. Tester loads the cube, hand-tests and reports a result. Everyone builds; rotate when the mentor calls time.

Evidence for the robot team

Our change: _____

In the hand test, the cube: _____

We predict this will help because: _____

Hand testing is preparation. The powered trials tell us whether it works on the robot. Record three runs on the team sheet. Success means the robot crosses the finish, keeps its cargo, stays in the lane and stops in the clear runout.

No hands in a moving robot's path. Wait for a full stop before resetting. Store your cradle with your team label for the showcase.